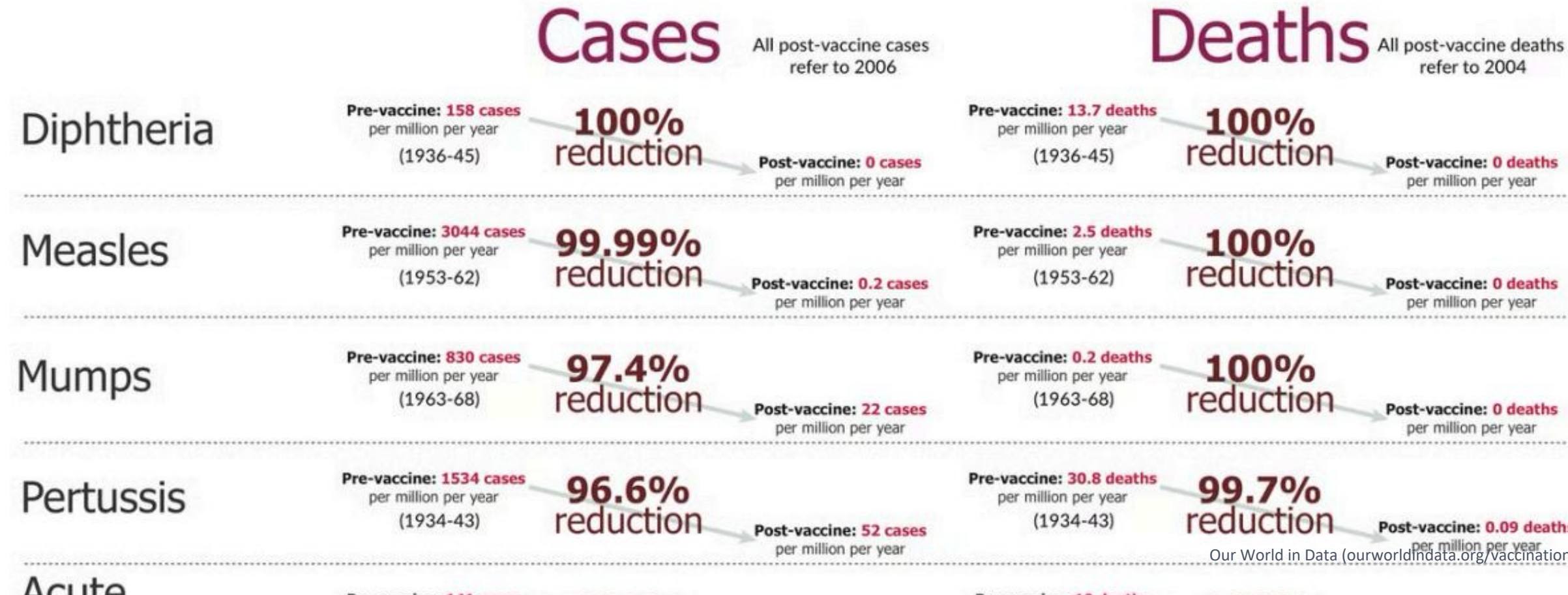
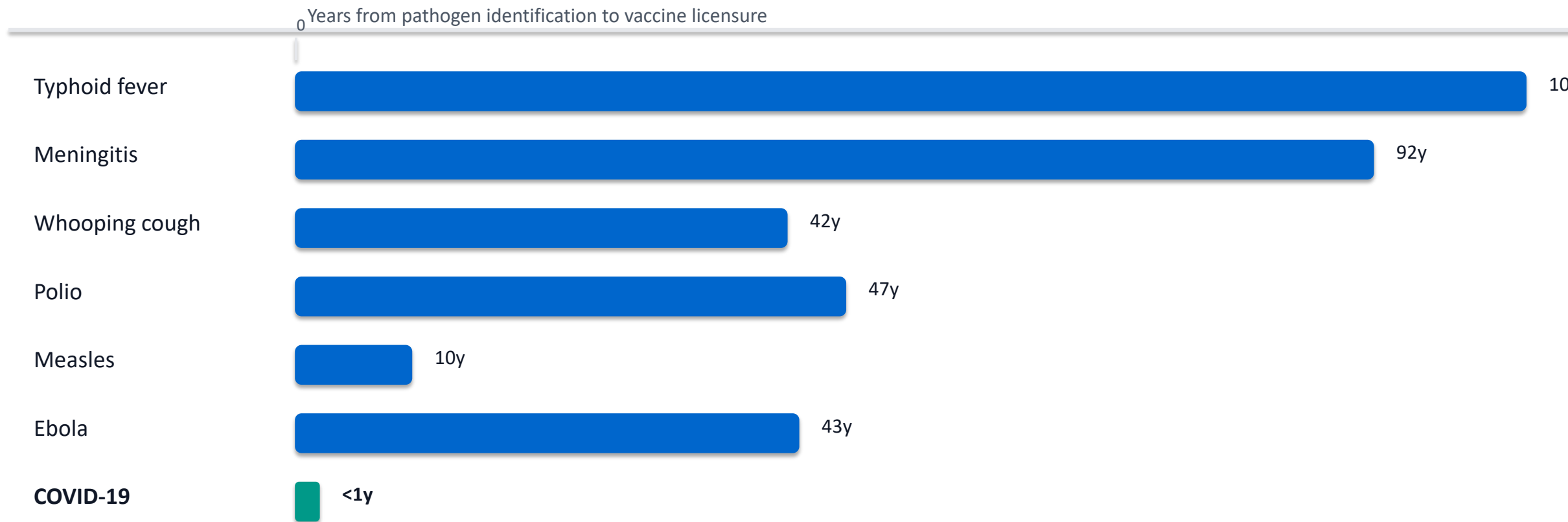


The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

1796–2020 | For public health professionals and policy analysts | Source: Our World in Data



From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



COVID-19 achieved licensure in 2020 in under 1 year; earlier vaccines often took decades.

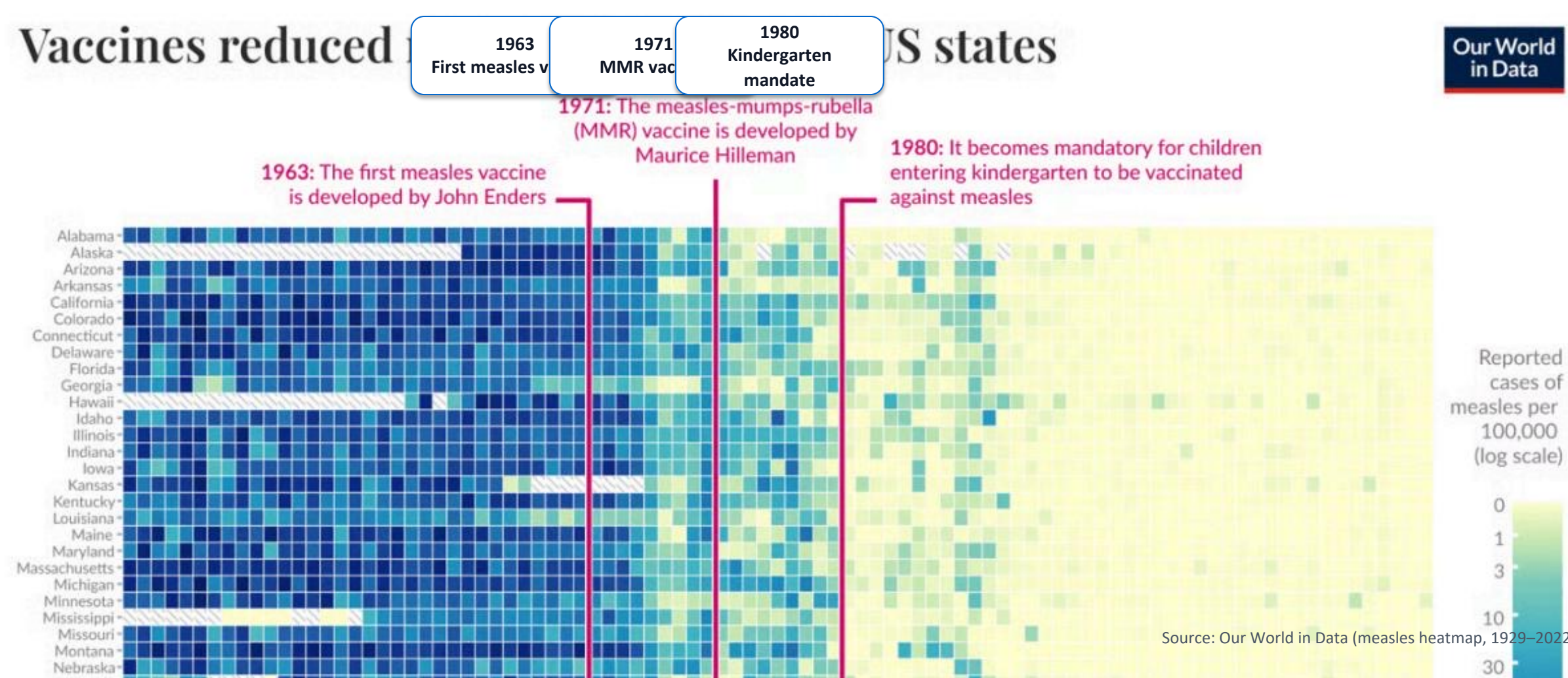
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-vaccine cases per million/yr	Post-vaccine cases per million/yr	Case reduction	Pre-vaccine deaths per million/yr	Death reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Rates shown per million people per year; reductions after vaccine introduction.

Source: Roush & Murphy (2007) via Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Smallpox: The Only Human Disease Eradicated Through Vaccination

1959: WHO passed a resolution to eradicate smallpox.

1967: WHO launched an intensified global eradication campaign.

1977: Last natural case occurred in Somalia.

1980: Smallpox declared eradicated worldwide (certified).

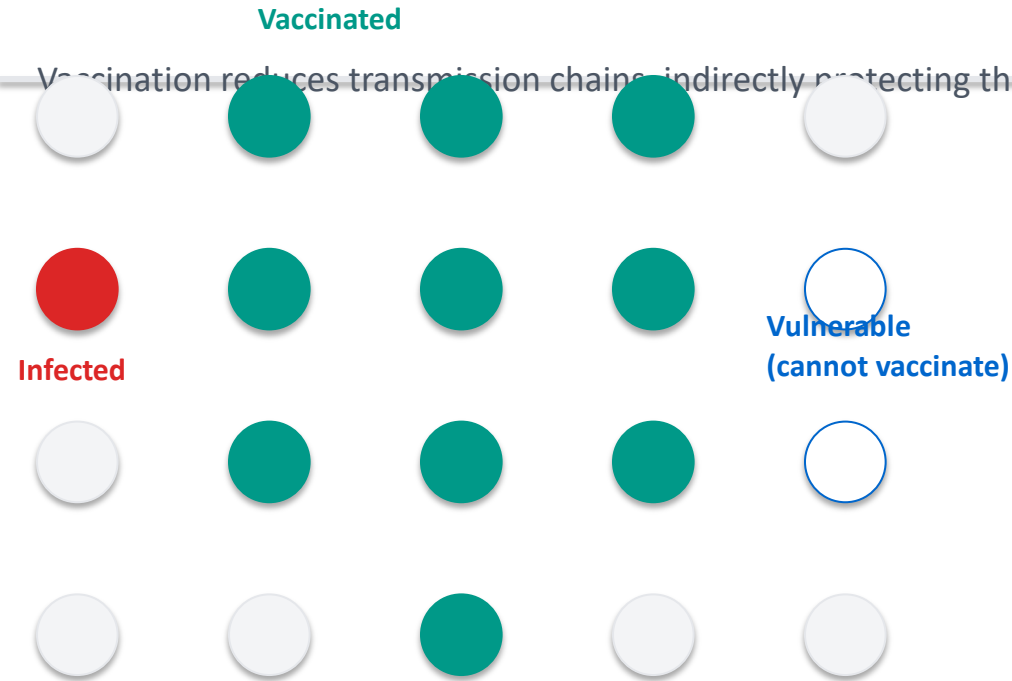
Why eradication was possible

- Infected only humans
- Symptoms were distinguishable
- Vaccine was cheap and effective

Vaccination innovation, from 1880 to 2020



Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Protects newborns, immunocompromised individuals, and cancer patients on chemotherapy.

Community protection depends on achieving sufficiently high coverage.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

40.5%

Pneumococcal case reduction

31.3%

Pneumococcal death reduction

89%

Hepatitis A case reduction

Acute hepatitis B: 83.9% case reduction (83.6% death reduction).

Globally: not everyone has access to vaccines; supply shortages put children at risk.

Poor scientific understanding can reduce trust and uptake.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and acute polio in the US.

Development timelines collapsed from 105 years (typhoid) to under 1 year (COVID-19).

Measles fell to near zero across all 50 states after the 1980 kindergarten mandate.

Herd immunity protects entire communities — especially people who cannot be vaccinated.

Gaps persist (e.g., pneumococcal disease), requiring sustained coverage, access, and trust.

Call to action: strengthen delivery systems, raise coverage, and protect trust — nationally and globally.